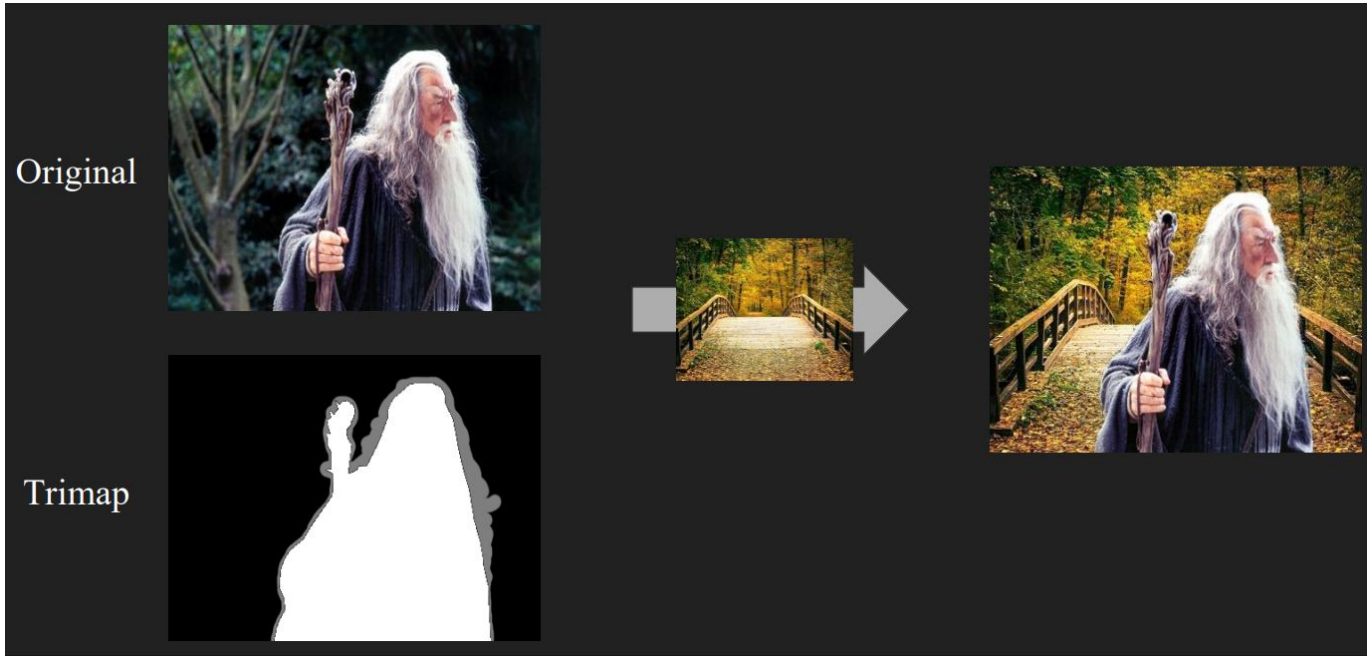
KNN Matting

Hw1

Why This Homework?

- VFX matting as an optimization problem
- Pixels as graph nodes



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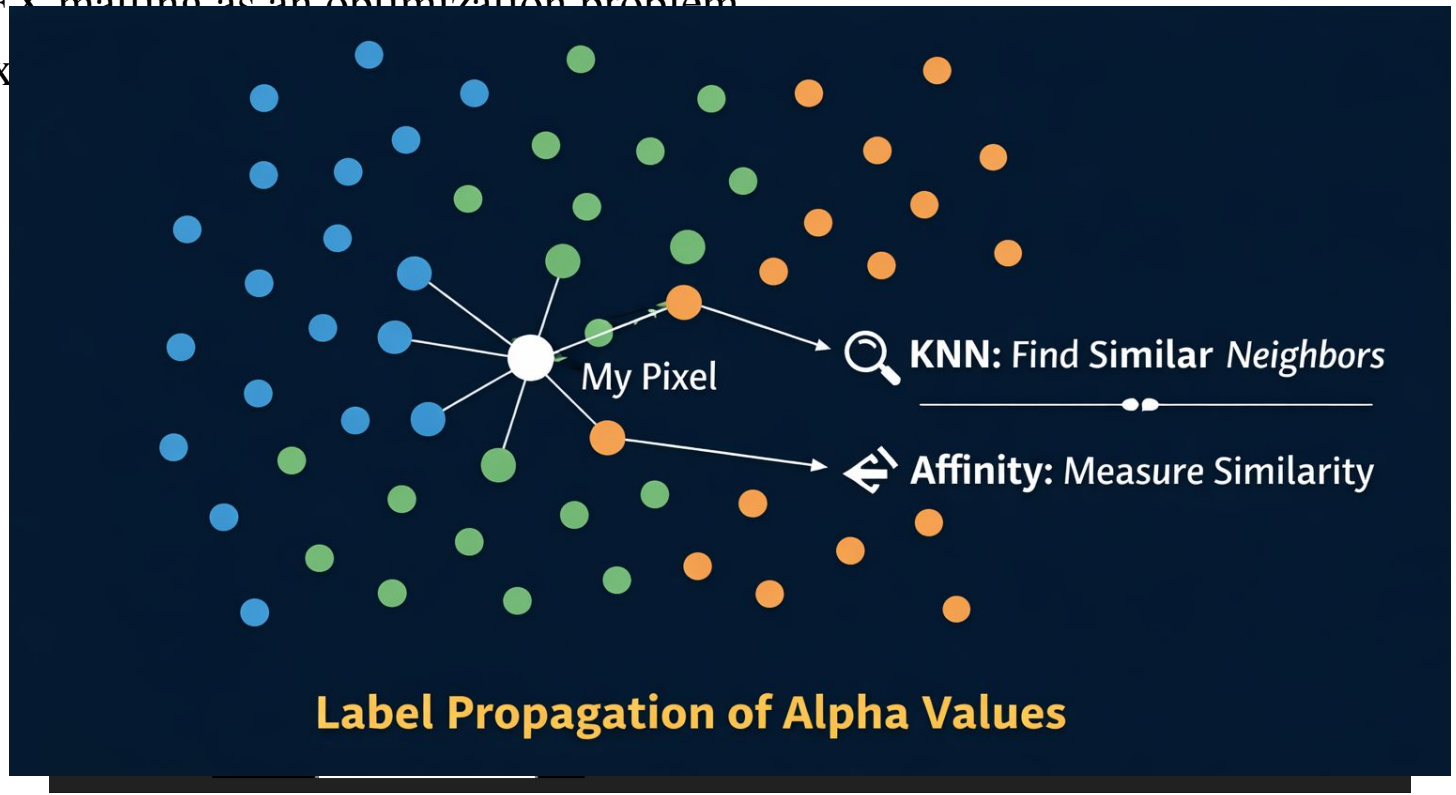
- Pix



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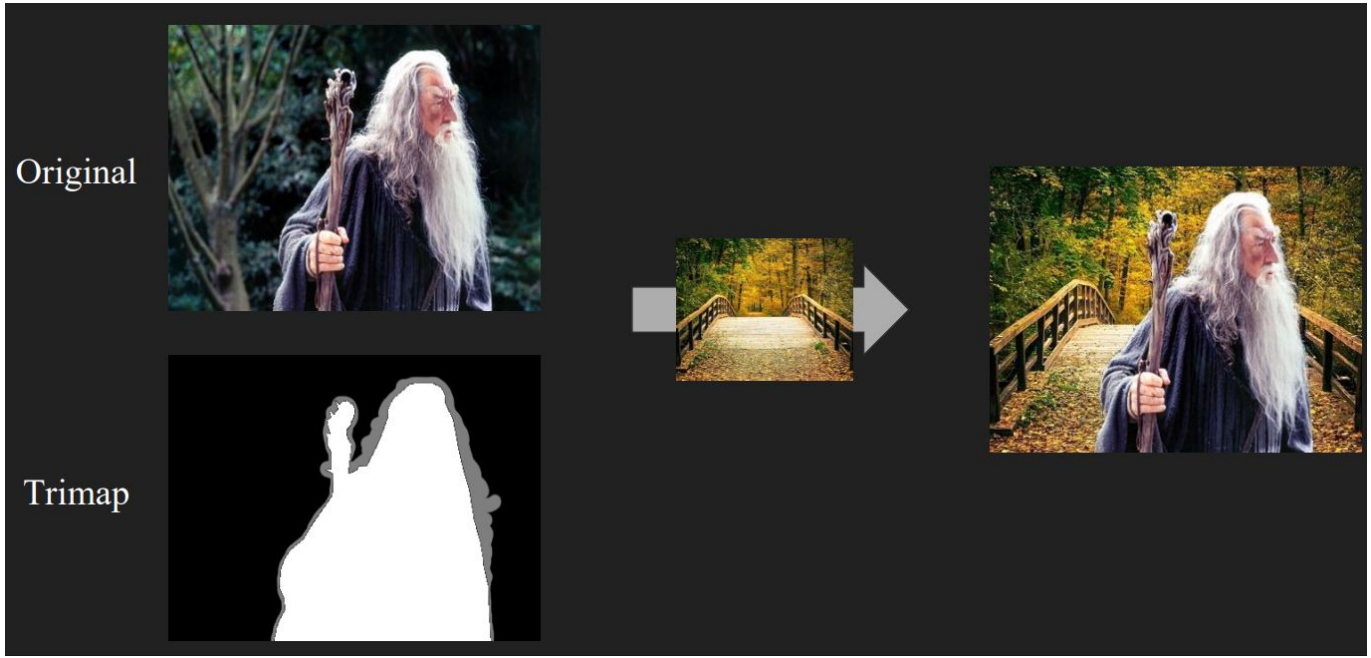
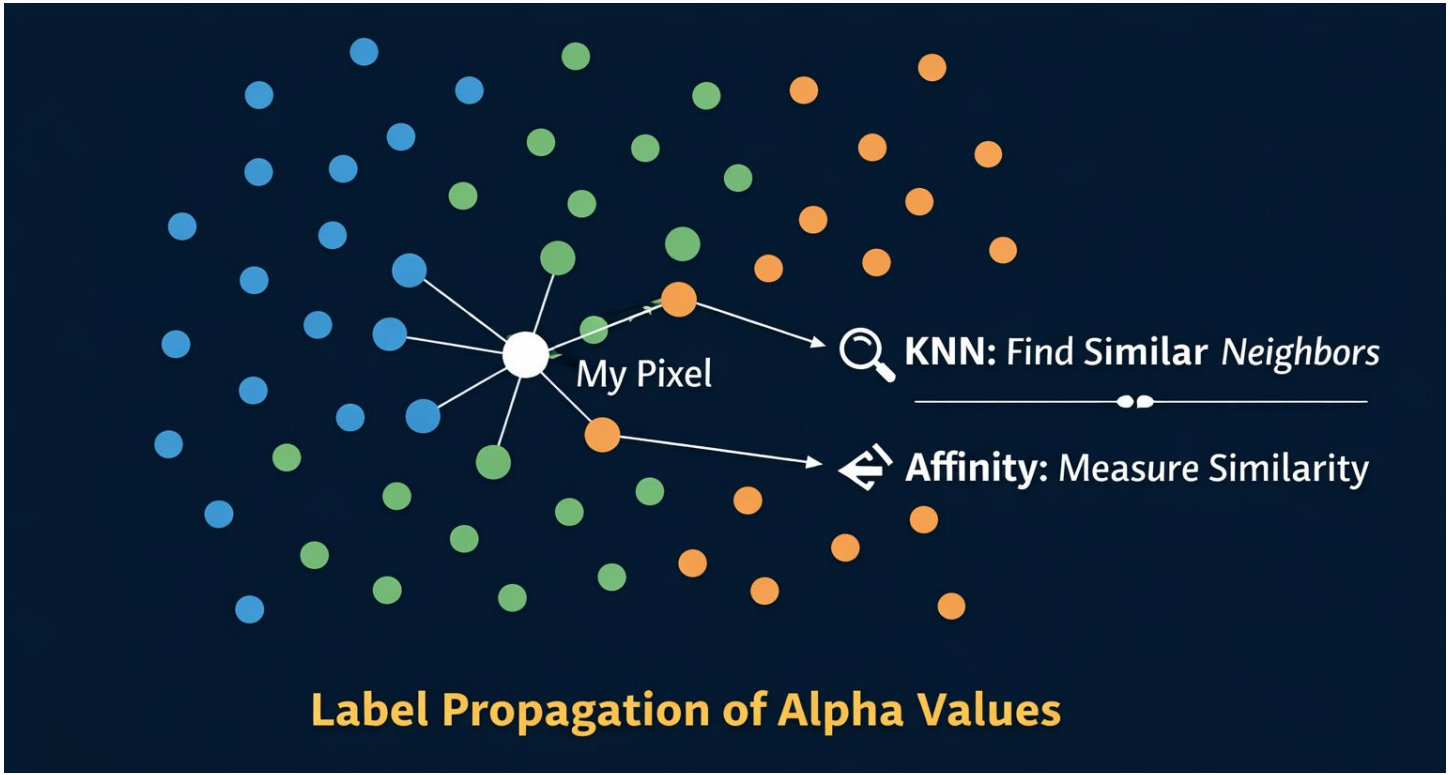


But How?

- By Trimap

Graph View


My Pixel
KNN: Find Similar Neighbors
Affinity: Measure Similarity
Label Propagation of Alpha Values

Overview

1. Find KNN and construct the affinity matrix A
    a. scikit-learn is allowed for finding KNN
    b. Try to play with the number of K and see how it affects the result
2. Solve for the linear system
    a. scipy and numpy is allowed for matrix computation
    b. Be careful with solving for the overdetermined system in case it has no solution
3. Composite some new images
    a. Pick a landscape photo on your own and try to attach your result onto it

Feature vector

- Baseline: [R, G, B], [R, G, B, x, y]
- Defines similarity

For [R, G, B], it is one of the key in KNN matting setting, “nonlocal” connections. It can get remote useful information without considering position.

Affinity Matrix A

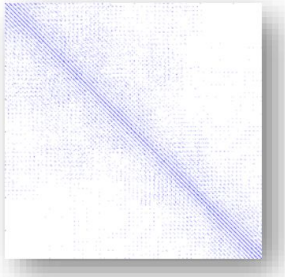
- Encodes pixel similarity
- **Sparse by KNN** → KNN is performed before computing A

Kernel function $k(i, j)$

$$k(i,j) = 1 - \frac{\|X(i) - X(j)\|}{C}$$

C is the least upper bound of $|X(i)-X(j)|$ to make $k(i, j) \in [0, 1]$.

$$A = \begin{bmatrix} k(1,1) & \cdots & k(1,n) \\ \vdots & \ddots & \vdots \\ k(n,1) & \cdots & k(n,n) \end{bmatrix}$$



A of KNN matting

Solve Linear System

- Balance non-local and user constraints

Evaluation for KNN matting (cont.)

- We want α of mark-up pixels fitting for users assignment.

$$\boldsymbol{M\alpha} \approx \boldsymbol{v}$$

$\boldsymbol{\alpha}$: the vector of all α
$\boldsymbol{M}$: diag($\boldsymbol{m}$), indicates whether a pixel is marked-up.
$\boldsymbol{v}$: *indicates the assigned alpha*

- Besides, $\boldsymbol{D\alpha} \approx \boldsymbol{A\alpha}$

$\boldsymbol{D}$: diag(D_i), an NxN diagonal matrix
$\boldsymbol{A}$: NxN affinity matrix composed of $k(i,j)$

- Our objective function becomes

$$Obj = (\boldsymbol{D} - \boldsymbol{A})\boldsymbol{\alpha} + \lambda(\boldsymbol{M\alpha} - \boldsymbol{v}) = (\boldsymbol{L} + \lambda\boldsymbol{M})\boldsymbol{\alpha} - \lambda\boldsymbol{v}$$

$\boldsymbol{L}$= ($\boldsymbol{D}$-$\boldsymbol{A}$)

λ: a weight controlling user's confidence on the markups

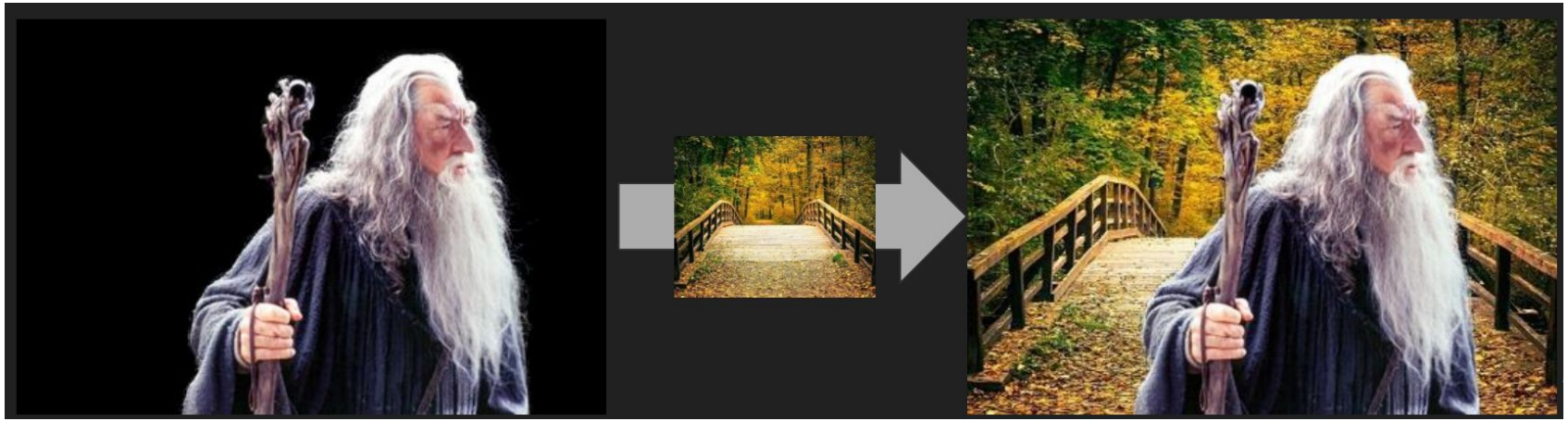
- The optimal $\boldsymbol{\alpha}$ can be estimated for the linear system.

This work can simultaneously extract multiple image layers, but we focus on fg and bg here.

Composition

- After getting F, B and α, we can simply change the background and composite a new image using the formula shown below.

$$C = \alpha F + (1 - \alpha) B$$



Requirements - Basic 80%

1. Feature Design - 10%
2. K Selection - 10%
3. Linear system Solver - 10%
4. Results - 50%
    a. Bear - 20%
    b. Woman - 10%
    c. White cloth - 20%

Requirements - Advanced 20%

- **Tackle white cloth / white wall problem - 10%**

  For this section, your score will be judged based on the outcome performance compared to your classmates.

- **Based on how much you do - 10%**

  - Solving for a large linear system is extremely time-comsuming, so find some other methods to speed it up and make a comparison on how long these methods take - 5%
  - Compare with other matting methods - 5%

  Your score will be judged by the effort you put into it.

Submission

- Deadline: 2026/4/3 23:59:59
  - Final score = original score * 0.9 for less than a week late (4/4 ~ 4/10)
  - Final score = original score * 0.8 for less than two week late (4/11 ~ 4/17)
  - and so on.
- Every submission should consist of the followings:
  - Your code (in python)
  - Report (.pdf) following the template
- Please put the above files into <your Student ID>_<your name>_HW1.zip and submit through E3.
- e.g.

  313551000_TA_HW1.zip
  |- main.py
  |- report.pdf

Reminder

- Please do not upload code that you have found on the internet.
- If you refer any code from GitHub or other open-source projects, you have to properly cite the source and comment on those code. Otherwise, you will get a penalty of 20 points or more.
- You should work on all the given images.